

Getting the Most Out of PostgreSQL

The amazing thing about the PostgreSQL RDBMS is that it has been developed entirely by a worldwide team of volunteers. It is not owned by any corporation or entity, and is totally free of cost. Its source code is available for anyone to modify and use as one pleases. Let's take a look at how to use it to our advantage.



PostgreSQL is a relational database management system (RDBMS). It is free and open source software, and its source code is available under the PostgreSQL licence, which is a liberal open source licence. Users are free to use, modify and distribute PostgreSQL in any form.

This article is logically divided into three sections. The first section is PostgreSQL specific and covers its installation, data types and DB related operations. The second section covers basic SQL, and the third discusses advanced SQL.

An overview of PostgreSQL and its salient features

PostgreSQL was designed to run on UNIX-like platforms. However, it was also designed to be portable so that it could run on various platforms such as Mac OS X, Solaris and Windows.

The salient features of PostgreSQL are:

- 1) Supports user-defined data types
- 2) Supports sophisticated locking mechanisms
- 3) Supports nested transactions (checkpoints)
- 4) Supports multi-version concurrency control
- 5) Supports asynchronous replication

Environment setup

You need to perform the following steps to set up the PostgreSQL development environment.

- 1) Visit <https://www.enterprisedb.com/downloads/postgres-postgresql-downloads> link.
- 2) Select the appropriate version and OS, and start downloading. I have downloaded 'postgres-10.1-3-linux-x64.run'.
- 3) Make the downloaded script executable and execute it as

a root user to begin installation:

```
$ chmod +x postgresql-10.1-3-linux-x64.run
$ sudo ./postgresql-10.1-3-linux-x64.run
```

- 4) Follow the on-screen instructions to complete the installation.

Basic database operations

In my environment, PostgreSQL is installed in the `/opt/PostgreSQL` directory. All its utilities are located under the `bin` directory. Include this directory under the `PATH` variable using the following command:

```
$ export PATH=$PATH:/opt/PostgreSQL/10/bin
```

Let us create a PostgreSQL database using the `createdb` command:

```
$ createdb -h localhost -p 5432 -U postgres testdb
```

In the above command:

- The `-h` option indicates the host's FQDN/IP address
- The `-p` option indicates port number
- The `-U` option indicates the database user
- `-testdb` indicates the database's name

 **Note:** Hereafter, we'll use the `psql` utility, which is the command line PostgreSQL client, to perform DB operations. `psql` supports SQL queries as well its own commands. All `psql` commands start with a backslash character—for instance, `\h`, `\q` and so on. Also, the `psql` prompt is denoted by `'postgres=#'` when it is not connected to any database, or `'testdb=#'` when it is connected to 'testdb'.